



ROBERT A. DENTON, INC.

2967 Waterview Dr., Rochester Hills, MI 48309

Tel (248) 852-5100 • Fax (248) 852-6060 • email: info@radenton.com • www.radenton.com



MODEL 1914A FEMUR LOAD CELL

HII 50M, HIII 95LM, HIII 50M, HIII 5F, THOR, BioSID, SID, EuroSID-1, SID-IIs and ES-2

Applications:

The Model 1914A Femur Load Cell was designed to provide a more comprehensive measurement of the forces acting upon the upper leg just above the knee joint. These measurements are useful for developing occupant protection systems in which "knee bolsters" and/or structural components are contacted by the dummy's legs during an impact. The Model 1914A Load Cell can be adapted to other applications by removing and/or replacing the end caps. The load cell contains six measurement channels: three forces and three moments. Each channel is a full bridge strain gage circuit. Two, 30-foot-long (9.15 m), twelve conductor, Teflon insulated, shielded cables exit the side of the load cell. One contains wiring for the force channels and the other contains wiring for the moment channels. With aluminum end caps (part #5632), the load cell can be used in the ES-2.

The Models 1914V and 1914 Femur Load Cells are identical to the Model 1914A except for the bridge resistance of each channel.

Specifications:

Channel	Capacity		Bridge Resistance (nom)			Output Sensitivity (nom)
	Metric	English	1914	1914A	1914V	
Fx	13344.7 N	3000 lb	350 ohms	350 ohms	120 ohms	1.8 mv/v
Fy	13344.7 N	3000 lb	350 ohms	350 ohms	120 ohms	1.8 mv/v
Fz	22241.1 N	3000 lb	480 ohms	700 ohms	120 ohms	1.2 mv/v
Mx	338.9 Nm	3000 in-lb	350 ohms	350 ohms	120 ohms	1.5 mv/v
My	338.9 Nm	3000 in-lb	350 ohms	350 ohms	120 ohms	1.5 mv/v
Mz	338.9 Nm	3000 in-lb	700 ohms	700 ohms	240 ohms	2.6 mv/v

Nonlinearity.....less than 1% of full scale
Hysteresis.....less than 1% of full scale
Crosstalk.....less than 5% of full scale
Excitation.....10 VDC nom. 15 VDC max.
Mass (load cell only).....(2.3 lb) 1.043 kg

Wiring:

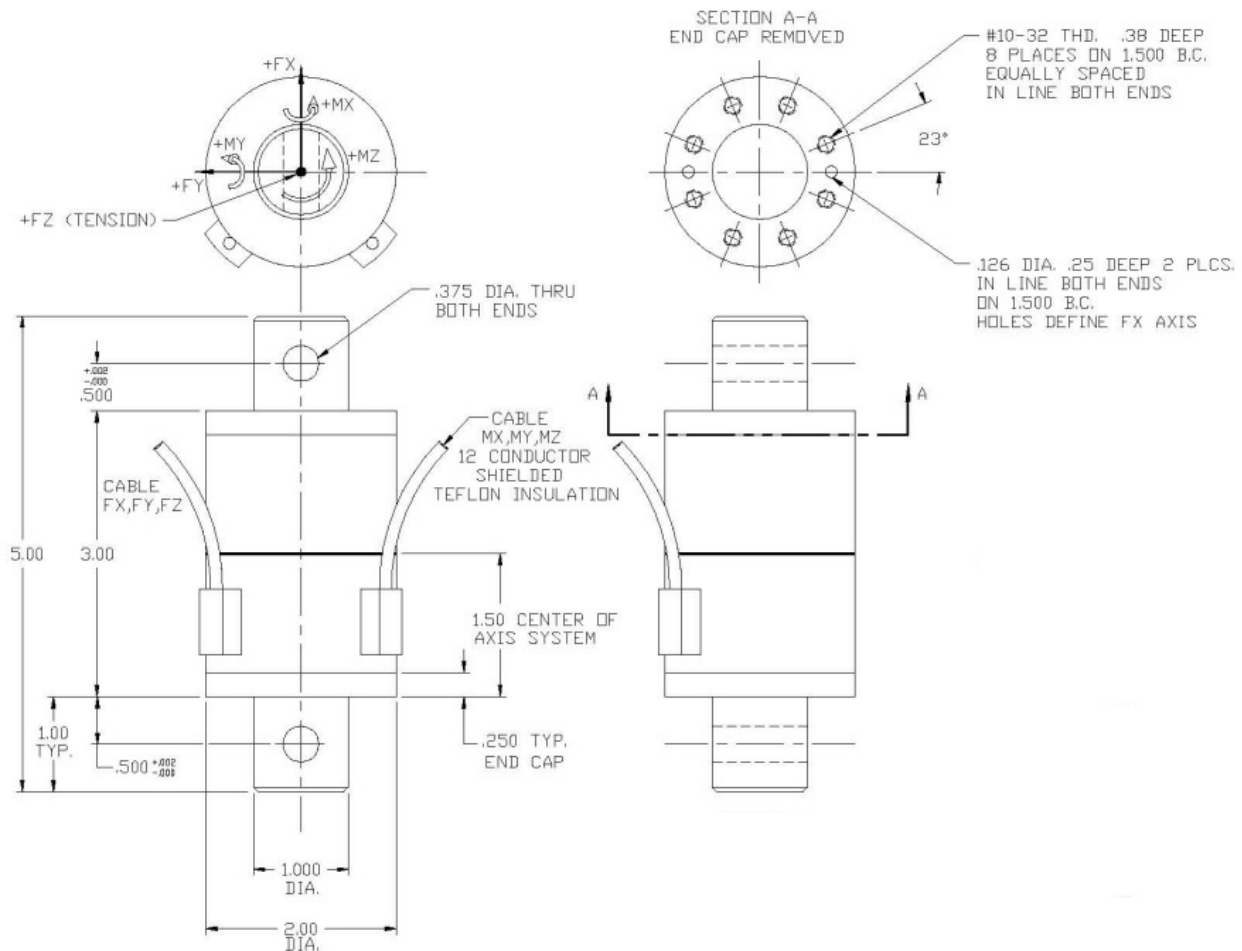
Fx & Mx Channels		Fy & My Channels		Fz & Mz Channels	
Brown	+Exc.	Red Stripe	+Exc.	Green	+Exc.
Red	+Sig.	Black	+Sig.	Blue	+Sig.
Orange	-Exc.	White	-Exc.	Violet	-Exc.
Yellow	-Sig.	Black Stripe	-Sig.	Gray	-Sig.



MODEL 1914 FEMUR LOAD CELL

NOTES

1. Use Denton Model 1957 calibration fixture. (Ref. Calibration Fixture drawing 1914-C)
2. If the model tag on the load cell includes a "J" suffix, the load cell is wired for SAE J211 Right Hand Rule sign convention.
3. If the model tag on the load cell includes a "G" suffix, the shield of the cable is grounded to the body of the part.
4. If the model tag on the load cell includes an "I" suffix, there is an ID module in the body of the load cell. The number following the "I" indicates the type of ID module.
5. The standard cable length for this load cell is 9.1 meters (30'). If the cable is cut shorter, the calibration will change. Upon request, the load cell can be made and calibrated with a custom cable length.



Creating the Standard in Safety Measurement Since 1974

Robert A. Denton, Inc. maintains a high level of control over the manufacture of its products to ensure quality, accuracy and reliability. Continued improvement necessitates that we reserve the right to modify these specifications without notice.

Robert A. Denton, Inc., 2967 Waterview Drive, Rochester Hills, Michigan 48309
 Tel: (248) 852-5100 • Fax: (248) 852-6060 • email: info@radenton.com
 Copyright © R. A. Denton, Inc., 2000